Boundary-Spanning Activity and Research and Development Management: A Comparative Study

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Abstract—Boundary-spanning activity (BSA) was studied in two departments of a large governmental research and development organization. BSA was found to be higher and to have significant effects for employees when departmental goals were unclear and technology was non-routine. Under these conditions, BSA was negatively related to role ambiguity and positively related to job satisfaction. Contrary to prior research, this study found roles with high levels of BSA to have favorable aspects for their incumbents depending upon goal clarity and type of technology of the parent organization. Implications for the management of research and development organizations are discussed.

INTRODUCTION

The research and development (R&D) organization is increasingly being viewed as an open system with boundaries that filter information into and out of an organization [1], [5], [18], [22]. As a means of facilitating the transfer of technology and information between organizations, boundary-spanning activity (BSA) has evolved; it may be defined as the interpersonal transfer of information across organizational boundaries. The importance of both formal and informal BSA to the effective management of R&D organizations has been confirmed by prior research [1], [2], [5], [14], [18].

BSA has also been found to transfer technology and information in informal ways through "invisible colleges" [3], [4], [6]. This body of research has shown that the diffusion of knowledge among scientists often occurs in ways not formally designed by the R&D organization; nevertheless, this form of BSA can be as important to the organization as BSA that is formally designated.

Much of the prior research into BSA found that designated boundary-spanning roles have negative aspects for their incumbents. Several researchers [8], [13], [14], [21] have found incumbents of boundary-spanning roles to experience such dysfunctions as lower job satisfaction, higher role conflict (inconsistent or conflicting expectations for role performance), and higher role ambiguity (vague or unclear expectations for role performance). The basic explanation for these negative role characteristics is that those in boundary-spanning roles must face conflicting expectations for performance from the different organizations with which they interface. In contrast, a non-boundary spanner, dealing only with the expectations for performance from his own organization, would face less conflicting and ambiguous expectations.

Given the importance of BSA for the effective functioning of R&D organizations, as well as the apparent negative aspects of boundary-spanning roles for their incumbents, there is clearly a need for research which will help R&D managers to design BSA and boundary-spanning roles which maximize effectiveness and minimize problems for incumbents. The purpose of this study, then, was to investigate BSA as well as key role and satisfaction variables of professional employees in two R&D departments which had different levels of goal clarity and different types of technology.

THEORETICAL FRAMEWORK AND HYPOTHESES

A theoretical framework (based on [9], [10], [20]) was developed which sees the clarity of organizational goals, the type of technology, and the degree of environmental uncertainty as determining the nature of information and technology transfer occurring between organizations. In turn, the nature of information to be transferred precedes and strongly affects the type of BSA. In addition, the nature of the information influences the importance and effects of BSA on role and satisfaction variables for employees. The diagram in Fig. 1 depicts these relationships.

Three basic hypotheses, which assume other factors are equal, were then generated from the theoretical framework: (a) BSA will be higher when organizational goals are unclear and technology is non-routine than when goals are clear and technology is routine; (b) role conflict and ambiguity will be higher when organizational goals are unclear and technology is routine; and (c) BSA will be more strongly related to role and satisfaction variables when organizational goals are unclear and technology is non-routine than when goals are clear and technology is routine.

The rationale for the first hypothesis is that the existence of unclear goals and non-routine technology would tend to increase the need for information from other organizations to help clarify goals and solve problems; hence, the amount of BSA would be higher. The basis for the second hypothesis is that the lack of goal clarity and a non-routine technology

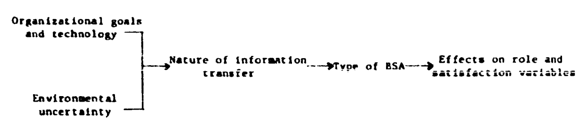

Fig. 1. Theoretical framework.

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would tend to cause higher role conflict and ambiguity since personnel would tend to be unsure about what is expected of them for role performance. The rationale for the third hypothesis is that BSA would tend to affect the roles and satisfaction of employees more strongly under unclear goals and non-routine technology because BSA would tend to be a more important activity for the organization than when goals are clear and technology is routine.

METHOD

This study was conducted in two departments of a large governmental R&D organization. Department *A* was selected because of its newness and lack of clear departmental goals. The department was created approximately one year before the survey and was charged with the "administration of all research and development activity" associated with an advanced, newly-developing technical field. Because of great environmental uncertainty, Department *A* lacked clearly stated goals as well as routine policies and procedures. Department *A* had a large number of external interfaces and boundary-spanning roles in addition to its many positions with primarily internal relations. The external interfaces of the department included over 20 universities, several large private contractors, and other governmental agencies, as well as six other departments in the R&D organization. Many of the employees of this department worked on tasks which were not related to their formal education and previous experience.

Questionnaires were distributed to the 88 professional employees of Department *A*. Fifty-eight questionnaires were returned (seven of these questionnaires were not usable). The actual sample of 51 respondents (respone rate = 58%) had an average age of 40 years, with a range of 29 to 57 years. All subjects had at least a bachelor's degree, with most having a graduate degree. Average total work experience was 17.5 years, and the average time in present position was 4.7 years.

Department *B* was selected for its similarity to Department *A* in all respects except goal clarity and routineness of technology. Department *B* provided complex computational and data processing support to approximately two dozen other departments in the R&D organization. The goals in Department *B* were clear and longstanding; support was provided through a series of complex but routine procedures. Departments *A* and *B* were similar in size, complexity, technical discipline, number of external interfaces, and employee biographical characteristics.

Seventy-seven of the 97 professional employees in Department *B* returned usable questionnaires (six unusable questionnaires were returned, resulting in a response rate of 79%). The average age of the subjects was 37 years, and the range was 25 to 55 years. All but three subjects in Department *B* had bachelor's degrees, while most had some graduate training. The average total work experience was 14.4 years, and the average time in present position was 5.1 years.

Measurements of Variables

Questionnaires were distributed to the subjects at their place of work and were returned directly to the researchers in postpaid envelopes. Responses were anonymous. A series of interviews was also conducted with the boundary spanners in Department *A* in order to complement and corroborate the questionnaire data.

Boundary-Spanning Activity (BSA): The level of BSA was measured by a four-item instrument which is included in the Appendix. The BSA scale was developed from items previously used and validated by Creighton, Jolly, and Denning [5]; scale items were similar to those successfully used by Allen and Cohen [1].

The BSA scale utilized in the present study concerned (a) sources of work-related information, and (b) contact with people outside and inside the R&D organization. The BSA score was obtained by summing the scores for each of the four questionnaire items which were measured on a five-point scale. To validate the BSA scale, the top management of each of the two departments was asked to identify the background characteristics of some employees which management thought to be high on BSA. Those identified as boundary spanners by top management in Department *A* all scored in the top third of their department's BSA scale distribution, and those identified as boundary spanners in Department *B* all scored in the top fourth of their department's BSA scale distribution. A *t*-test between the means of BSA scores of the identified boundary spanners and the other employees, moreover, was significant at the .001 level for both departments. Reliabilities (Coefficient Alpha) were .73 in Department *A*, and .76 in Department *B*. Correlation matrices for the BSA items are shown in the Appendix.

Role Conflict and Ambiguity: When expectations for behavior and performance are in conflict or inconsistent, role conflict can be experienced by an individual. Similarly, vague or unclear expectations can lead to role ambiguity. The two scales which Rizzo, House, and Lirtzman [17] developed and validated were used to measure role conflict and ambiguity. There were eight items in the role conflict scale and six items in the role ambiguity scale. The items in the role ambiguity scale were reverse scored since they were worded in the direction of clarity. The following seven-point scale was used for each item of the role conflict and ambiguity scales:

Very False						Very True
1	2	3	4	5	6	7

Job Satisfaction: The Job Description Index which was developed and extensively validated by Smith, Kendall, and Hulin [19] was used to measure five job satisfaction dimensions: satisfaction with the work itself, co-workers, supervision, pay, and opportunities for promotion.

RESULTS

A series of *t*-tests was run on the scores from the two departments for BSA, role conflict and ambiguity, and the five dimensions of job satisfaction. Table I presents the results of this analysis. A correlational analysis was also conducted on the above scores for each of the two departments, and differences between the correlations for the departments were also

TABLE I
A Comparison of Two R&D Departments on Boundary-Spanning Activity and Role and Job Satisfaction Variables

Variable		Department A (N=51)	Department B (N=77)	Differences (Two-tailed tests)
Boundary-Spanning Activity	mean	11.73	9.16	t=3.95, p<.001
	S.D.	3.98	3.27	
Role Conflict	mean	37.57	32.17	t=3.53, p .001
	S.D.	7.72	8.88	
Role Ambiguity[a]	mean	24.47	22.61	t=1.83, p<.08
	S.D.	6.76	7.53	
Work Satisfaction	mean	32.57	31.77	t=.39, N.S.
	S.D.	10.51	11.81	
Supervision Satisfaction	mean	34.94	37.58	t=1.17 N.S.
	S.D.	12.18	12.57	
Co-Worker Satisfaction	mean	39.31	39.68	t=.18 N.S.
	S.D.	11.05	11.07	
Promotion Satisfaction	mean	5.73	8.21	t=1.94, p<.05
	S.D.	6.47	7.39	
Pay Satisfaction	mean	21.12	20.79	t=.34 N.S.
	S.D.	5.38	5.42	

[a]Role ambiguity items were reverse scored since they were worded in the direction of clarity.

TABLE II
Correlations Between Boundary-Spanning Activity and Role and Job Satisfaction Variables

Variable	Department A (N=51)	Department B (N=77)	Differences Between Correlations (Two-tailed tests)
Role Conflict	.08	.03	z=.27 N.S.
Role Ambiguity[a]	-.31[b]	.02	z=1.84 p<.07
Work Satisfaction	.24	.09	z=.81 N.S.
Supervision Satisfaction	-.11	.14	z=1.36 N.S.
Co-Worker Satisfaction	.41[c]	.09	z=1.87 p<.06
Promotion Satisfaction	.43[c]	.04	z=2.27 p<.02
Pay Satisfaction	.28[b]	-.20	z=2.65 p<.01

[a]Role ambiguity items were reverse scored since they were worded in the direction of clarity.
[b]$p < 0.05$.
[c]$p < 0.01$.

tested. The results of the correlational analyses are reported in Table II.

The data in Table I show Department A to be higher than Department B on BSA, role conflict, and role ambiguity. These data reflect the basic differences between the departments on goal clarity and technology. Department B has a higher score on satisfaction with opportunities for promotion, but the other four dimensions of job satisfaction are not significantly different.

The correlations in Table II show BSA to be significantly related to lower levels of role ambiguity, and higher levels of satisfaction with co-workers, pay, and opportunities for promotion in the non-routine technology and environment of Department A. BSA has low and non-significant correlations with role and satisfaction variables in the routine technology and environment of Department B. There are significant differences between the correlations of the two departments in this analysis for BSA and role ambiguity and BSA and satisfaction with co-workers, pay, and opportunities for promotion.

In summary, then, Department A had higher levels of BSA, role conflict and ambiguity, and lower satisfaction with opportunities for promotion than Department B. BSA, moreover, was significantly related to role and satisfaction variables in Department A, but not in Department B.

DISCUSSION AND IMPLICATIONS FOR R&D MANAGEMENT

The results of this study basically show that in Department A, where goals were unclear and technology non-routine, BSA was quite an important variable for explaining role ambiguity and job satisfaction. BSA, however, was not important for explaining these variables in Department B where goals were clear and technology routine. In addition, the needs for the transfer of non-routine information and technology in Department A were associated with a significantly higher level of BSA than in Department B where information and technology transfer were of a more routine nature.

The research hypotheses, then, were supported by the results. An inference from the data is that BSA can be quite important in an R&D organization when effective organizational performance requires non-routine information and technology from other organizations in order to solve problems and accomplish tasks. In this situation, BSA can also help to clarify goals for the parent organization since boundary spanners can learn the expectations of outsiders for the performance of the parent organization.

Those personnel who engage in a high level of BSA are also provided with an important power resource—information. Pettigrew [15], for example, found that boundary spanners can use their centrality in information flows to obtain better rewards in the form of pay, promotions, and favorable relations with colleagues. Boundary spanners, moreover, can reduce ambiguity for fellow workers and hence enhance job satisfaction as described by House [7]. These advantages for incumbents of boundary-spanning roles, when goals are unclear and technology non-routine, can help to explain why BSA was significantly associated with lower role ambiguity and higher job satisfaction in Department A, and not in Department B.

Some important implications for R&D management can also be drawn from the present study. To utilize BSA properly, R&D management should first assess the goals and technology of its organization. This assessment can help give management an indication of the nature of BSA in the organization in terms of the relationship between the organizations being spanned and the type of information and technology being transferred.

When goals are unclear, technology non-routine, and interorganizational relations include high levels of non-routine information transfer, BSA will tend to be of substantial importance to effective organizational functioning. Under these conditions, R&D management should put highly effective personnel into roles requiring a high level of BSA. Prior work [11], [12], [16], & [23] has concluded that effective boundary-spanners tend to be flexible, pragmatic, and able to tolerate conflicting and ambiguous role expectations. These roles, moreover, can often be used for the development of managerial talent. Under the described conditions, those with a high level of BSA were found to have less role ambiguity and higher job satisfaction; therefore, roles with high levels of BSA can be quite desirable. Personnel in these roles also tend to be visible and well-known both inside and outside their organizations, and often have a high degree of contact with top management due to their sources of information [15]. As a result, favorable career opportunities and access to organizational rewards are often related to a high degree of BSA.

CONCLUSION

As environments become more dynamic and the ability of organizations to monitor changes and transfer information and technology across their boundaries becomes increasingly important, the effective utilization of BSA in R&D organizations becomes critical for successful performance. BSA, after all, is the essential link between an organization and its environment.

Prior research has found some negative characteristics of boundary-spanning roles. This study, however, found that when organizational goals are unclear and technology is nonroutine, those in positions with high levels of BSA can benefit from their roles. Role ambiguity was found to be lower and job satisfaction higher for those with high levels of BSA; moreover, the visibility and access to important information of these roles can be used to enhance career opportunities. Under these conditions, R&D management would do well to put talented personnel into roles with high levels of BSA for career and management development.

APPENDIX

The following four items were used to measure boundary-spanning activity. Each item had a five-point scale of "0," "1-2," "3-4," "5-6," and "More than 6" (Table III).

Instructions: In this section you are asked to report on sources of *work-related* information that you use or lead others to use. "Work-related" should be broadly interpreted. For example, both *Scientific American* and *The Wall Street Journal* might be used to keep informed, although specific application to a work task might be unusual.

1. Indicate the *total* number of magazines, journals, and newspapers which you regularly read.

2. During the last month, what was the frequency with which you recommended specific information sources (e.g., journal article, knowledgeable colleague, report, magazine, etc.) to a colleague?

TABLE III

Boundary-Spanning Activity Scale Item Correlation Matrices

Department A					Department B				
Item	1	2	3	4	Item	1	2	3	4
1		.37	.28	.23	1		.34	.35	.33
2			.41	.45	2			.54	.50
3				.65	3				.59

3. During the last month, indicate the frequency with which *you have sought* information/advice from members of other organizations outside (name of organization).

4. During the last month, indicate the frequency with which members of organizations outside (name of organization) have sought information/advice *from* you.

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